Saumya Maurya

SUMMARY

Data Engineer with 3+ years of experience in designing and optimizing large-scale data pipelines, cloud architectures, and analytics platforms across AWS, Azure, and GCP. Skilled in ETL/ELT design, Spark-based processing, SQL optimization, and data modeling. Strong background in data governance, automation, and performance tuning, enabling reliable and scalable data-driven insights for business teams.

SKILLS

- **Programming:** Python, SQL, PySpark, Shell Scripting
- **Big Data & Cloud Platforms:** AWS (Glue, Redshift, Lambda, S3), Azure (Data Factory, Synapse Analytics, Data Lake Gen2), GCP (BigQuery, Dataflow, Cloud Storage)
- **Data Engineering Tools:** Apache Spark, Kafka, Apache Beam, Airflow
- **ETL & Data Integration:** AWS Glue, Azure Data Factory, SSIS, Apache NiFi
- **Data Wrangling & Transformation:** Pandas, PyArrow, Spark SQL, Window Functions
- **Business Intelligence:** Tableau, Power BI, Amazon QuickSight
- **Metadata & Governance:** Azure Purview, AWS Glue Data Catalog
- **DevOps & Automation:** Git, Jenkins, Docker, AWS Lambda, GCP Cloud Functions
- **Machine Learning Integration:** Vertex AI, Google Document AI
- **Database Optimization:** Redshift, Synapse, SQL Server, PostgreSQL

WORK EXPERIENCE

Data Engineer **Jun 2024 - Present**
Bliss Ladders, ON

- Partnered with data architects and BI analysts to modernize data warehouse architecture for BI and analytics; consolidated multi-source data (CSV, Parquet, APIs) into a centralized AWS-based lakehouse.
- Migrated legacy ETL jobs to serverless pipelines using AWS Glue, Apache Spark, and Redshift Spectrum, improving data processing throughput by 55% and enabling real-time analytics.
- Developed data transformation scripts using Python (pandas, pyarrow) and PySpark for high-volume datasets, cutting transformation runtime by 40% through parallel processing and optimized memory handling.
- Tuned SQL queries and implemented partitioning, indexing, and distribution strategies in Redshift, reducing query latency by 60% and improving dashboard refresh times significantly.
- Redesigned QuickSight dashboards and OLAP cubes, enhancing report performance by 50% through pre-aggregated metrics, semantic modeling, and efficient query execution.
- Implemented AWS Glue Data Catalog and Lambda-based automation for metadata management, ensuring complete data lineage, schema evolution, and governance compliance across all environments.

Associate Data Engineer **Jan 2021 – Feb 2023**
CMC Ltd., India

- Designed and implemented data aggregation pipelines using Azure Data Factory, Apache Spark, and Kafka, enabling ingestion and transformation of multi-source performance data with a 60% efficiency boost.
- Developed automated validation and schema mapping scripts in Python, improving data accuracy by 35% and reducing exception-handling effort by 50% across financial reporting workflows.
- Built optimized SQL and Spark queries in Azure Synapse Analytics, leveraging CTEs, window functions, and dynamic partitions to handle 5B+ record datasets with improved reliability and query speed (45%).
- Partnered with BI and Data Science teams to implement PySpark-based unit tests and data quality gates, cutting data discrepancies in reporting pipelines by 70%.
- Delivered interactive Tableau and Power BI dashboards for fund performance tracking, increasing visibility into KPIs by 50% and enabling faster investment decisions.
- Implemented Azure Purview for metadata and lineage tracking, ensuring 90% traceability coverage and full alignment with audit and compliance standards.

EDUCATION

Postgraduate Diploma in Artificial Intelligence **May 2024 – Dec 2024**
Georgian College, Barrie (GPA – 4.0)

Postgraduate Diploma in Mobile Application Development **May 2023 – Dec 2023**
Georgian College, Barrie (GPA – 4.0)

Bachelor of Computer Applications
Maharaj Sayajirao University, Vadodara, Gujarat, India

CERTIFICATION

- AWS Academy Machine Learning Foundations (2024)
- AWS Academy Cloud Foundations (2024)
- Python Programming – Great Learning Academy (2020**)**

ACADEMIC PROJECTS

Heart Guard: Predicting Cardiovascular Risk

- **Objective:** Predicted heart disease likelihood using clinical data (UCI Heart Disease Dataset).
- **Approach:** Conducted Exploratory Data Analysis (EDA) to identify key features like age, cholesterol, and blood pressure.
- **Results:**
- Achieved 92% accuracy using a Random Forest model optimized via TPOT AutoML.
- Enabled early detection and improved patient outcomes by providing actionable and efficient insights for healthcare professionals.

Customer Churn Prediction for Telecom Industry

- **Objective:** Predicted customer churn using demographic and usage data to inform retention strategies.
- **Approach:** Designed and executed models, including Neural Networks (78.75% accuracy) and CNNs (78.68%), aligning with business requirements for performance metrics.
- **Key Contributions:** Identified critical features like contract type and tenure, enabling targeted interventions that improved retention.